FIVE-CHAPTER INTERVIEW REFERENCE

DATA SCIENCE FOUNDATIONS

The Interview Field Guide

Confident explanations. Defensible calculations.
Production judgment that goes beyond the formula.

DESCRIBE › MODEL › INFER › DECIDE

89	08	02
Interview Q&As	Production cases	Mock rounds

For Data Science · Applied ML · Product Experimentation
Built around one consistent YouTube-scale example

REFERENCE MAP

Inside this guide

Start with the interview ladder, then study any chapter independently. Every topic includes the explanation, assumptions, and failure modes worth remembering.

01 **Descriptive Statistics**
Summaries, spread, outliers, correlation

02 **Probability Foundations**
Bayes, conditional probability, expectation

03 **Probability Distributions**
Bernoulli, Binomial, Poisson, Normal, Exponential

04 **Statistical Inference**
Sampling, CLT, SE, CI, sample size

05 **Hypothesis Testing & A/B Testing**
Power, t-tests, experiment design, false discoveries

QUICK TOOLS & PRACTICE

CROSS-CHAPTER — Tough Interview Scenarios

FORMULA SHEET — The One-page Whiteboard Reference

COMMON INTERVIEW TRAPS — 25 Rapid Corrections

MOCK INTERVIEW — Two Short Rounds

FINAL REVISION PLAN — Three Passes

What makes this interview-focused?
Answers are organized to help you say the right thing out loud: definition, intuition, assumptions, calculation, and what can fail at production scale.

Scope: Chapters 1–5 | **Focus:** Data Science, Applied ML, experimentation, product analytics | **Example:** YouTube-scale recommendation platform

> **How to use this guide:** For each question, first give a 20–40 second answer out loud. Then use the **Interview answer** and **Probe** to check if you can defend it under follow-up questioning. Calculations are deliberately small enough to reproduce on a whiteboard. This is a revision reference, not a substitute for hands-on analysis.

The 60-second mental model

- **Descriptive statistics:** What happened in the data? Which summary is reliable?
- **Probability:** How likely are events, and how does new evidence update our beliefs?
- **Distributions:** What mechanism generated the outcomes, and what uncertainty should we expect?
- **Statistical inference:** What can sample data tell us about the population?
- **Hypothesis testing and A/B testing:** Is an apparent effect convincing, and should we act on it?

Interview answering framework

Define → Explain intuition → Name assumptions → Work one small example → Mention failure mode → Make a decision.

For a senior-level answer, saying the formula is rarely enough. Explain what would break its assumptions at 10 million users, what you would monitor, and how the result changes a product decision.

CHAPTER 1 — Descriptive Statistics

Interview objective: Summarize messy, skewed, large-scale user behavior without hiding important structure.

Fast recall: the vocabulary

Concept	Interview-ready meaning	YouTube example
Mean	Sum of values / count	Average daily watch minutes
Median	50th percentile	Typical user's watch time
Mode	Most frequent value or category	Most popular session-length bucket
Variance	Average squared deviation from a mean	Dispersion in minutes squared
Standard deviation	Spread in original units	Watch-time variability in minutes
Percentiles	Position in the distribution	p90 user watches at most X minutes
IQR	Q3 − Q1	Spread of middle 50%
Skewness	Distribution asymmetry	Long right tail from heavy users
Correlation	Strength/direction of association	Sessions and watch time

Core interview questions

Q1. When would you use mean rather than median?

Interview answer: The mean uses every observation and is directly useful for totals, resource planning, and additive outcomes. The median describes the typical observation and is resistant to extreme values. At YouTube, a few marathon viewers may raise mean watch time while leaving the median almost unchanged. Report both if the distribution is skewed.

Probe: If total watch minutes are the revenue-relevant quantity, can the median replace the mean? **No** — the median doesn't determine the aggregate sum.

Q2. Explain how an outlier affects mean, median, and mode.

Interview answer: An extreme high value increases the mean, often has little effect on the median, and usually doesn't alter the mode unless it changes the most frequent value. Consider watch times [10, 15, 20, 25, 30] with mean 20 and median 20. Replace 30 with 300: the mean becomes 74, while the median remains 20.

Probe: Do you always delete outliers? **No.** Determine whether they're errors, fraud, rare genuine users, or a separate population.

Q3. What is the difference between population and sample variance?

Interview answer: Population variance averages squared deviations using denominator N. The usual unbiased estimator of population variance from an IID sample uses denominator n−1, because the sample mean is estimated from those observations. The denominator adjustment is Bessel's correction; it does not mean the sample variance of a particular sample must equal the population variance.

$$\text{Population: } \sigma^2 = \Sigma(x_i - \mu)^2 / N \quad | \quad \text{Sample: } s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

Q4. Why do we square deviations in variance?

Interview answer: Raw deviations from the mean sum to zero. Squaring prevents positive and negative deviations from cancelling and gives greater weight to large departures. The resulting units are squared, so SD takes the square root to return to the original units. Other dispersion measures, such as mean absolute deviation, use absolute values instead.

Q5. Explain p50, p90, p95, and p99 to a product manager.

Interview answer: p90 is the value at or below which roughly 90% of observed values fall, depending on the empirical quantile convention. For daily watch time, p50 describes a typical user, p95 identifies the higher-use tail, and p99 shows extreme usage. Report them alongside the observation period, eligibility rules, and whether the distribution is user- or event-weighted.

Probe: Does p95 = 120 minutes mean exactly 5% watch 120 minutes? **No.** Roughly 5% are above that percentile; ties can alter exact proportions.

Q6. How do quartiles and IQR help identify outliers?

Interview answer: Q1 is the 25th percentile, Q3 the 75th, and IQR = Q3 − Q1. Tukey's descriptive fences are Q1 − 1.5×IQR and Q3 + 1.5×IQR. Values outside them are flagged for investigation, not automatically deemed invalid. The rule can over-flag extremes in heavy-tailed distributions.

Q7. What is skewness, and how do you spot it?

Interview answer: Right skew has a long upper tail, common for spend, watch time, and session counts. Often mean > median, though this ordering isn't a universal mathematical rule. Inspect a histogram and quantiles, not just a skewness number. A log transform can help visualize positive skew but changes interpretability.

Q8. What is kurtosis? Is it just 'peakedness'?

Interview answer: Kurtosis is based on the standardized fourth moment and is especially sensitive to extreme deviations; describing it only as peakedness is misleading. High kurtosis signals unusually influential tails relative to a Normal benchmark, so averages and ordinary uncertainty estimates may require careful diagnostics.

Q9. Explain a z-score and its limitation.

Interview answer: $z = (x - \mu)/\sigma$ says how many SDs an observation lies from the mean. If mean watch time is 50 and SD is 10, then 70 minutes has $z = 2$. A z-score standardizes distance; it does not by itself assign a probability unless a distributional model is justified.

$$z = (x - \mu) / \sigma$$

Q10. What's the difference between covariance and correlation?

Interview answer: Covariance reflects whether two variables vary together but depends on their units and scale. Pearson correlation divides covariance by both SDs, making it dimensionless between −1 and +1. Correlation near zero rules out strong *linear* association, not every nonlinear relationship.

$$\rho(X,Y) = \text{Cov}(X,Y) / (\sigma_x \, \sigma_y)$$

Q11. Pearson vs Spearman: when would you choose each?

Interview answer: Pearson measures linear association between numerical variables. Spearman is Pearson correlation on ranks and measures monotonic association; it's often less sensitive to extreme magnitudes and useful for ordered data. Inspect scatterplots before treating either coefficient as complete evidence.

Q12. Why doesn't correlation imply causation?

Interview answer: Two metrics can move together because of a third factor, reverse causation, selection effects, or coincidence. For example, watch time and likes correlate partly because highly active users do more of both. To claim the recommendation algorithm *caused* an increase, prefer random assignment and a clearly defined experimental estimand.

Q13. What is Simpson's paradox?

Interview answer: An overall association may reverse after stratifying by relevant groups because group composition differs. If B has more heavy viewers, its overall watch time can appear higher even when A performs better within each viewer segment. Always inspect segmentation and treatment-assignment balance.

Q14. What is a weighted mean, and when can weighting go wrong?

Interview answer: A weighted mean is $\Sigma(w_i x_i)/\Sigma w_i$. For 80% casual users averaging 30 minutes and 20% heavy users averaging 120, the population mean is 0.8×30 + 0.2×120 = **48 minutes**. Wrong weights — for example, using observed response share when responses aren't representative — can produce misleading estimates.

Q15. What summaries would you report for 10 million daily watch-time observations?

Interview answer: Define the unit first (per user per day), then report sample size, missingness, zeros, mean, median, SD, p25/p75/p90/p99, distributions by region/device/tenure, and anomalies. Compare trends across consistent time windows. Avoid mixing per-session and per-user means: highly active users otherwise receive disproportionate weight.

Whiteboard drill — descriptive statistics

Data: [10, 20, 30, 40, 100]. Mean = **40**; median = **30**; population variance = **1,000**; population SD ≈ **31.62**; Q1/Q3 depend on the quartile convention. A high-tail user pulls the mean above the median.

Interview trap: Different libraries use different quantile interpolation and SD denominator conventions. State the method before comparing reported results.

CHAPTER 2 — Probability Foundations

Interview objective: Reason correctly about uncertainty, dependencies, conditionals, rare events, and base rates.

Fast recall: event rules

Rule	Expression	Why it matters
Complement	$P(A^c) = 1 - P(A)$	At least one event = 1 − none
Addition	$P(A \cup B) = P(A)+P(B)-P(A \cap B)$	Avoid double-counting overlap
Conditional	$P(A \mid B) = P(A \cap B)/P(B)$	Update probability given evidence
Multiplication	$P(A \cap B) = P(A \mid B)P(B)$	Sequential/joint outcomes
Independence	$P(A \cap B) = P(A)P(B)$	Factorization condition
Total probability	$P(A) = \sum_i P(A \mid B_i)P(B_i)$	Combine mutually exclusive cases

Core interview questions

Q1. What is a probability space?

Interview answer: It specifies possible outcomes (sample space), measurable events, and a probability assignment satisfying non-negativity, total probability 1, and countable additivity for disjoint events. In product analysis, define the experiment and unit: 'a randomly chosen user clicks within 24 hours' is different from 'a randomly chosen impression gets clicked.'

Q2. Mutually exclusive vs independent — are they the same?

Interview answer: No. Mutually exclusive events cannot happen together: $P(A \cap B) = 0$. Independent events satisfy $P(A \cap B) = P(A)P(B)$. Two nonzero-probability mutually exclusive events are necessarily dependent. For example, an impression being *only* in treatment A or *only* in treatment B is mutually exclusive for that assignment.

Q3. If A and B overlap, how do you calculate P(A or B)?

Interview answer: Add their probabilities and subtract the overlap once: $P(A \cup B) = P(A)+P(B)-P(A \cap B)$. If 40% like a video, 30% share it, and 10% do both, then 60% do at least one.

Q4. What is conditional probability?

Interview answer: It is the probability of A restricted to cases where B has occurred: $P(A \mid B)=P(A \cap B)/P(B)$ for $P(B)>0$. 'Probability a user likes a video given they finished it' is different from the overall like probability; the conditioning group changes the denominator.

Q5. Explain Bayes' theorem intuitively.

Interview answer: Bayes combines how likely evidence is under a hypothesis (likelihood) with how common the hypothesis was before seeing evidence (prior). The result is the updated conditional probability (posterior). It is especially important for rare-event detection, because even a good signal can generate many false alarms when the underlying event is rare.

$$P(H|E) = P(E|H)\,P(H) / P(E)$$

Q6. Walk through a rare-event recommendation-quality alert.

Interview answer: Suppose 1% of video recommendations are truly problematic. A detection system flags 90% of problematic cases (sensitivity 0.90) and falsely flags 5% of good cases (FPR 0.05). For 10,000 recommendations: 100 problematic → 90 true flags; 9,900 good → 495 false flags. Among 585 flags, only **90/585 ≈ 15.38%** are truly problematic. This is the base-rate effect.

Q7. What's the law of total probability?

Interview answer: Break the population into exhaustive, nonoverlapping groups and average conditional probabilities using their group shares. If 80% are casual users with 10% click probability, and 20% are heavy users with 30% click probability, overall click probability is 0.8×0.1+0.2×0.3 = **14%**.

Q8. Does 'A makes B more likely' prove A causes B?

Interview answer: No. $P(B|A) > P(B)$ is an association, potentially driven by self-selection, confounders, or reverse causality. Causal claims require additional design or assumptions; randomized assignment is valuable because it helps break confounding links between group assignment and potential outcomes.

Q9. What is expected value?

Interview answer: The probability-weighted average of a random variable's possible values. For 80% casual users at 30 minutes and 20% heavy at 120, E[X] = 0.8×30+0.2×120 = **48 minutes**. Expected value describes a long-run average; it need not be a possible individual outcome.

$$\text{Discrete: } E[X] = \Sigma_x\, x \cdot P(X=x)$$

Q10. How do you calculate variance from expectations?

Interview answer: $\text{Var}(X)=E[(X-\mu)^2]=E[X^2]-E[X]^2$. For the two-point watch-time population, E[X]=48 and $E[X^2]=0.8\times900+0.2\times14{,}400=3{,}600$, so $\text{Var}(X)=3{,}600-2{,}304=$**1,296 min²** and SD=**36 min**.

Q11. Why is linearity of expectation useful?

Interview answer: $E[X+Y]=E[X]+E[Y]$ even when X and Y are dependent, provided expectations exist. For user watch time, expected total minutes across users is the sum of expected minutes per user. Independence becomes essential for certain *variance* simplifications, not expectation additivity.

Q12. Can you add variances for any two variables?

Interview answer: Generally $\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)+2\text{Cov}(X,Y)$. If X and Y are independent, covariance is zero and variances add. Positive correlation makes the total more variable; negative covariance can make it less variable.

Q13. Joint, marginal, and conditional probabilities: what's the difference?

Interview answer: Joint P(A,B) describes both events. Marginal P(A) sums or integrates over the other variable. Conditional P(A|B) rescales joint probability by P(B). In a two-way table of device type and watch completion, read joint cells first, then row/column totals and within-group percentages.

Q14. What is calibration in probabilistic classification?

Interview answer: Among events predicted at probability about 0.8, roughly 80% should occur over an appropriate evaluation set. Calibration is distinct from ranking ability: a model can rank perfectly but assign badly calibrated probabilities. Check calibration across time and important segments, especially after data or base-rate shifts.

Q15. What is the probability of at least one rare failure across repeated trials?

Interview answer: Under independent trials with per-trial failure probability p, $P(\text{at least one in } n \text{ trials}) = 1-(1-p)^n$. With p=0.01 across 100 independent requests, this is about **63.4%**. If failures share a common cause, independence is inappropriate and the result changes.

Q16. How do conditional independence assumptions affect ML?

Interview answer: Conditional independence says variables are independent after conditioning on specified information; it is stronger than merely having zero correlation. Naive Bayes assumes feature conditional independence given the class, which often fails literally but can still yield useful predictions. Dependencies can make probabilistic confidence unreliable.

Whiteboard drill — base-rate trap

A classifier with 95% sensitivity and 95% specificity screens an event with 1% prevalence. For 10,000 observations: true positives ≈ 95; false positives ≈ 495; precision ≈ **95/(95+495)=16.1%**. **Interview insight:** Accuracy and sensitivity alone do not determine precision; prevalence matters.

CHAPTER 3 — Probability Distributions

Interview objective: Select an appropriate probability model, interpret its parameters, and recognize assumption failures.

Fast recall: how to choose a distribution

Question	Candidate	Support / key parameter
Did the user click?	Bernoulli	0 or 1; p
How many clicks in fixed n trials?	Binomial	0…n; n, p
How many arrivals in an interval?	Poisson	0,1,2,…; λ
Where does a roughly symmetric continuous measurement fall?	Normal	Real line; μ, σ^2
Time until next event under constant hazard	Exponential	t ≥ 0; rate λ

Core interview questions

Q1. PMF vs PDF vs CDF?

Interview answer: A PMF assigns probability to discrete values: P(X=k). A PDF represents density for continuous variables, so interval probability comes from area under the curve, not the density at one point. A CDF gives cumulative probability F(x)=P(X≤x) for either type.

Q2. Why isn't a PDF value itself a probability?

Interview answer: For a continuous variable, P(X=x)=0 at a single point under an ordinary continuous density, even though f(x) can be positive or greater than 1. Probability is $\int_a^b f(x)dx$ over a range. A density can exceed 1 when concentrated on a sufficiently narrow interval.

Q3. How do you get a tail probability from the CDF?

Interview answer: F(x)=P(X≤x), so P(X>x)=1−F(x). For a continuous distribution, P(X≥x)=1−F(x) as well because point masses are zero. For a discrete distribution, strict versus non-strict inequalities require attention to the mass at x.

Q4. What is Bernoulli, and what are its mean and variance?

Interview answer: Bernoulli describes one binary trial, X∈{0,1}, where P(X=1)=p. Mean E[X]=p and variance p(1−p). It is appropriate for a single eligible user's click indicator, not for the total clicks across many trials.

> $$\text{Bernoulli: } P(X=x) = p^x(1-p)^{1-x} \quad | \quad E[X]=p \quad | \quad \text{Var}(X)=p(1-p)$$

Q5. When is Binomial appropriate?

Interview answer: X counts successes in a fixed number n of independent Bernoulli trials with the same success probability p. $P(X=k)=C(n,k)p^k(1-p)^{n-k}$. If user click probabilities differ widely or events are correlated, an ordinary Binomial model may underestimate variance.

Q6. Calculate a Binomial probability.

Interview answer: With n=10 independent users and p=0.2 probability of clicking, exactly two clicks has probability $C(10,2)(0.2)^2(0.8)^8 = 45\times0.04\times0.16777216 \approx$ **0.302**. Mean clicks = np = 2; variance = np(1−p)=1.6.

Q7. When is Poisson useful?

Interview answer: Poisson models counts of events over exposure intervals under a constant event rate with independent increments in a Poisson process. For λ=3 arrivals per minute, $P(X=0)=e^{-3}\approx$ **0.0498**. A Poisson variable has mean and variance both equal to λ; strongly overdispersed counts suggest another model or unmodeled heterogeneity.

Q8. Binomial versus Poisson: what's the distinction?

Interview answer: Binomial counts successes from a fixed number of trials; Poisson counts events during a fixed exposure at a given rate. A Binomial(n,p) can be approximated by Poisson(np) when n is large and p small. Do not treat every event-count process as Poisson without checking assumptions.

Q9. What are the defining properties of a Normal distribution?

Interview answer: It is symmetric and bell-shaped, determined by mean μ and variance σ^2. For a Normal distribution, the mean, median, and mode coincide; about 68%, 95%, and 99.7% lie within 1, 2, and 3 SDs. Watch time itself is often non-Normal, but averaged outcomes can be approximately Normal under the CLT.

Q10. How would you estimate P(X>54) if X~Normal(42,12²)?

Interview answer: Standardize: z=(54−42)/12=1. The upper tail beyond z=1 is approximately **15.87%**. This relies on the Normal model; an observed p99 far beyond its Normal prediction would indicate a heavy-tail mismatch.

Q11. What does the Exponential distribution model?

Interview answer: Waiting time until the next event in a homogeneous Poisson process. With event rate λ per unit time, E[T]=1/λ, $F(t)=1-e^{-\lambda t}$, and $P(T>t)=e^{-\lambda t}$. It assumes a constant hazard; time-of-day traffic patterns often violate this assumption.

Q12. Explain exponential memorylessness.

Interview answer: $P(T>s+t \mid T>s)=P(T>t)$. If waiting time is Exponential, having waited s units does not change the distribution of the additional wait. Many product processes, including churn and long user sessions, are not memoryless because behavior changes with elapsed time.

Q13. How are Poisson and Exponential connected?

Interview answer: In a homogeneous Poisson process with rate λ, event counts over time t follow Poisson(λt), while the time to the next event follows Exponential(λ). They're two perspectives on the same process: counts in a window and waiting time until the next arrival.

Q14. How do you know whether your fitted distribution is wrong?

Interview answer: Compare support, observed mean–variance relationship, histograms, quantiles, tails, zero frequency, and group-level residual patterns. For daily watch time, zeros, long right tails, and user heterogeneity may make a single Normal or Exponential model unsuitable. Use predictive checks, not just a single goodness-of-fit score.

Q15. Why is expected value sometimes a poor description of an individual outcome?

Interview answer: E[X] can lie between modes or fail to correspond to any possible outcome. In our two-point population, E[X]=48 minutes even though an individual watches either 30 or 120. Always distinguish the mathematical expectation from a 'typical user' and report quantiles where appropriate.

Whiteboard drill — choose the model

One like indicator → Bernoulli. **Number of likes among 20 independent opportunities** → Binomial if p is stable. **Number of arrivals in a fixed minute** → Poisson if its process assumptions hold. **Wait until next arrival** → Exponential for a homogeneous Poisson process. **Average watch time across many independent users** → often an approximately Normal *sampling distribution*, even when individual watch times are skewed.

CHAPTER 4 — Statistical Inference

Interview objective: Translate a sample into a defensible population estimate; quantify uncertainty without mistaking precision for validity.

Fast recall: population versus sample

Population	Sample or estimator	Meaning
N	n	Size of population / sample
μ	$\bar{x}$	Population mean / observed sample mean
σ^2	s^2	Population variance / sample variance
σ	s	Population SD / sample SD
p	$\hat{p}$	Population proportion / sample proportion

Core interview questions

Q1. Population vs sample: what's the interview distinction?

Interview answer: The population is the set we want to make claims about; the sample is the observed subset. At YouTube, 'all eligible active users this week' is a population definition, while the randomly selected 1,000 users are a sample. If the target excludes new users or includes bots, that choice changes the meaning of the estimate.

Q2. Parameter, statistic, estimator, estimate — differentiate all four.

Interview answer: Parameter: fixed but often unknown population quantity μ. Statistic: numerical function of sample data. Estimator: the rule $\bar{x} = \Sigma X_i/n$ viewed as random before sampling. Estimate: realized result, say $\bar{x} = 42$ minutes. An estimate need not equal the parameter.

Q3. What is a sampling distribution?

Interview answer: The distribution of a statistic across repeated samples of the same design and size. If we repeatedly sample 1,000 users and calculate the mean, the histogram of those sample means approximates its sampling distribution. Don't confuse it with the distribution of individual watch times.

Q4. Explain the Central Limit Theorem correctly.

Interview answer: For IID observations with finite variance, the standardized sample mean approaches a Standard Normal distribution as sample size grows. This does not make original observations Normal and does not remove selection bias. For strong skew/heavy tails, large samples may still be required for a good approximation.

$$(\bar{x} - \mu) / (\sigma / \sqrt{n}) \to N(0,1) \quad \text{as } n \text{ increases, under CLT assumptions}$$

Q5. Difference between the Law of Large Numbers and CLT?

Interview answer: LLN explains convergence of the sample mean to μ as n grows under its conditions. CLT describes the limiting distribution of standardized fluctuations around μ. LLN says *where the mean goes*; CLT says *how the sampling error behaves*.

Q6. What is Standard Error, and why is it different from SD?

Interview answer: SD measures dispersion of individual observations; SE measures dispersion of a statistic across repeated samples. For an IID sample mean, SE=σ/√n, estimated by s/√n when σ is unknown. A sample with SE 2 minutes doesn't guarantee its realized mean is within 2 minutes of truth.

$$\text{SE}(\bar{x}) = \sigma / \sqrt{n} \quad | \quad \text{Estimated SE}(\bar{x}) = s / \sqrt{n}$$

Q7. Can you derive why SE decreases as √n rather than n?

Interview answer: For independent observations with variance σ^2, $\text{Var}(\Sigma X_i)=n\sigma^2$. Dividing the sum by n to form $\bar{x}$ divides variance by n^2: $\text{Var}(\bar{x})=\sigma^2/n$. Taking the square root yields SE=σ/√n. Quadrupling n halves SE; doubling n reduces SE by a factor of √2, not 2.

Q8. What is sampling error versus Standard Error?

Interview answer: Actual sampling error is $\bar{x}-\mu$ for one realized sample; SE is the SD of $\bar{x}$ across possible samples. If μ=50, $\bar{x}$=54, and SE=2, the realized error is +4 minutes while SE is still 2. These answer different questions.

Q9. What is an unbiased estimator?

Interview answer: One whose expectation equals the target parameter across repeated samples: $E[\hat{\theta}]=\theta$. For a simple random sampling design, the sample mean is unbiased for the population mean. Unbiased doesn't mean an individual estimate is correct; low variance, robustness, and MSE also matter.

Q10. What do consistency and MSE tell you?

Interview answer: Consistency means an estimator approaches the parameter as the sample grows under suitable conditions. Mean squared error measures expected squared estimation error: MSE=variance+bias^2. A small-bias estimator can sometimes have lower MSE than an unbiased estimator with high variance.

Q11. How do you interpret a 95% confidence interval?

Interview answer: The interval-building procedure covers the fixed population parameter in about 95% of repetitions under the model and sampling assumptions. Once an interval has been computed, frequentist confidence is not the probability that the fixed parameter 'moves inside' it. Do not confuse CI for a mean with a prediction interval for an individual.

Q12. Calculate a 95% CI for a mean.

Interview answer: Suppose $\bar{x}$=50 minutes and estimated SE=3 minutes, with a justified Normal approximation. Margin of error ≈1.96×3=5.88; CI ≈**[44.12, 55.88]**. If the sample is small and σ unknown, a t critical value is generally more suitable.

$$\text{Approximate 95\% CI: } \bar{x} \pm 1.96 \times \text{SE}$$

Q13. What is margin of error? What changes it?

Interview answer: For a symmetric CI, margin of error is the distance from its center to either endpoint. Under a Normal mean interval, MOE=$z^*\sigma/\sqrt{n}$. Higher confidence means a larger critical value and wider CI; larger independent sample size reduces MOE. It says nothing by itself about selection bias.

Q14. Calculate sample size for a desired CI width.

Interview answer: For 95% confidence, expected population SD σ=30 minutes, and desired MOE E=3, $n=(1.96\times30/3)^2=384.16$; **round up to 385**. Halving E to 1.5 raises the calculation fourfold, to n=1,537 after rounding. This is precision planning for a mean, not power planning for an A/B effect.

> $n = (z \times \sigma / E)^2$ (large-population, simple random/independent-sample approximation)

Q15. Why might a confidence interval be narrow and completely wrong?

Interview answer: If 70% of the sample are heavy viewers but only 20% of the population are heavy viewers, the estimate targets the wrong mixture. With casual at 30 and heavy at 120 minutes, population mean is 48 but this sample mixture averages 93. More data from the same biased source can tighten an invalid interval.

Q16. What happens if users contribute many correlated events?

Interview answer: The number of rows is not the effective number of independent units. A million view events from 10,000 users can produce overconfident uncertainty if events are treated as IID. Use the randomized unit as the analysis unit when appropriate, or employ cluster-robust variance, resampling by user, or hierarchical methods.

Q17. When would you use a bootstrap confidence interval?

Interview answer: Bootstrap methods approximate the sampling distribution by resampling observed units. They can help when an analytical SE is difficult to derive, but the resampling scheme must respect the data-generating process. Resample users rather than individual views when dependence is at user level; bootstrap cannot repair an unrepresentative original sample by itself.

Q18. If a sample is 1,000 out of 10 million, should you use finite population correction?

Interview answer: Under simple random sampling without replacement, SE includes a correction roughly $\sqrt{((N-n)/(N-1))}$ with σ defined over the finite population. For 1,000 out of 10 million the factor is almost 1, so it's negligible. It matters when a large fraction of the finite population is sampled.

Whiteboard drill — sampling bias versus sampling uncertainty

True population: 80% casual users ×30 minutes + 20% heavy users ×120 minutes = **48 minutes**. Convenience sample: 30% casual ×30 + 70% heavy ×120 = **93 minutes**. A confidence interval of [92.7, 93.3] might be narrow around the wrong target. **Interview conclusion:** Ask how people entered the sample before interpreting the interval.

CHAPTER 5 — Hypothesis Testing & A/B Testing

Interview objective: Design, analyze, and critique an experiment without confusing a small p-value with a meaningful causal win.

Fast recall: the decisions

Statistical term	Correct interpretation	Common wrong interpretation
H_0	Starting null claim	'Proven true' unless rejected
α	Prechosen Type I error bound	Probability H_0 is true
p-value	Tail probability under null model	Probability that H_0 is true
Type I error	Reject true H_0	Missing a real effect
Type II error	Fail to reject false H_0	Detecting an effect that isn't real
Power	P(reject H_0 given specified true effect)	Probability an effect exists
Confidence interval	Uncertainty about effect/parameter	Range containing 95% of users

Core interview questions

Q1. State H_0 and H_1 when YouTube wants to know whether B improves watch time.

Interview answer: Let $\delta=\mu_B-\mu_A$, where μ values represent the relevant population means under each algorithm. For a prespecified one-sided improvement test, H_0: $\delta\leq 0$ versus H_1: $\delta>0$. For any change, H_0: $\delta=0$ versus H_1: $\delta\neq 0$. Choose the question before seeing data.

Q2. What is a significance level?

Interview answer: Alpha is the maximum Type I error probability the test is designed to control under the null, if model and analysis assumptions hold. A level-0.05 test limits false rejection probability to at most 5% for each prespecified valid test. Alpha is not a 5% probability that H_0 is true.

Q3. Explain a test statistic to a product manager.

Interview answer: It measures how far our observed result is from the null reference relative to the uncertainty in that result. If B−A=3 minutes and SE of the difference is 1.5 minutes, the standardized statistic is (3−0)/1.5=**2**. Larger absolute values typically indicate less compatibility with a zero-difference null model.

Q4. What is a p-value precisely?

Interview answer: Under a specified null hypothesis and valid model, a p-value is the probability of a statistic at least as extreme as the observed statistic in the directions defined by the test. A one-sided Z=2 gives $p\approx 0.0228$. It is not the chance that the null is true, the lift percentage, or the probability of replication.

Q5. What's wrong with saying 'p = 0.03 means 97% chance B is better'?

Interview answer: That's the inverse probability fallacy. The p-value is calculated assuming the null and measures how surprising the test result is, not $P(H_1 | \text{data})$. It also says nothing directly about effect size or practical value. Report estimated lift, interval, design validity, and costs along with p.

Q6. Explain Type I and Type II errors using deployment decisions.

Interview answer: Type I: B has no true improvement but the test rejects H_0, leading to a false win. Type II: B really improves watch time but the test fails to reject H_0, causing us to miss a win. The cost of a false rollout and the cost of missing an improvement both affect design choices.

Q7. What is statistical power?

Interview answer: For a particular true effect and test design, power is the probability that the test rejects H_0. It equals $1-\beta$. An experiment with 80% power to detect a **2-minute** genuine improvement would detect it in about 80% of repetitions where that is the true effect. Power is not universal across effect sizes.

Q8. What affects statistical power?

Interview answer: Increasing independent sample size, increasing the true effect magnitude in the tested direction, and reducing noise generally increase power; using a stricter alpha generally reduces it, holding other factors fixed. Allocation ratio, correlation, cluster structure, and analysis method also affect power.

Q9. One-tailed versus two-tailed: how do you choose?

Interview answer: A one-sided test looks for a prespecified directional change; a two-sided test looks for either increase or decrease. At $\alpha=0.05$ under Normal assumptions, critical values are +1.645 for right-tailed and ±1.96 for two-tailed tests. Switching to one-sided after seeing a favorable direction invalidates the intended error-rate interpretation.

Q10. Z-test vs t-test: what's the actual distinction?

Interview answer: A Z reference is appropriate when the standardization has a known Normal null distribution or sound large-sample approximation. A t-test explicitly accounts for estimating variance, particularly with small samples under its assumptions. It's not simply 'n below 30 means t, otherwise Z'. For independent A/B groups with unknown potentially unequal variances, Welch's t-test is a common default.

Q11. When should you use paired t-test instead of Welch's independent t-test?

Interview answer: Paired t-tests operate on within-pair differences, such as the same users measured before/after under a design where pairing is valid. Welch compares means of independent groups without assuming equal variances. Randomized A/B users assigned once to either A or B form independent groups, not natural pairs.

Q12. Outline an end-to-end trustworthy A/B test.

Interview answer: Define target user population and randomization unit; predeclare primary metric and guardrails; plan detectable effect, alpha, power, and sample size; randomize consistently; validate assignment and event logging; analyze at the correct unit; calculate effect, interval, and p-value using the planned approach; examine guardrails and rollout risks. Don't equate statistical significance with automatic launch.

Q13. Walk through a numerical YouTube A/B test.

Interview answer: Suppose $n_A=n_B=1{,}600$, $\bar{x}_A=42$, $\bar{x}_B=44$, and $s_A=s_B=20$ minutes. Difference $\hat{\delta}=2$. Welch $SE=\sqrt{20^2/1600+20^2/1600}=\sqrt{0.5}\approx 0.707$. $t\approx 2/0.707=$ **2.83**; two-sided $p\approx$ **0.0047**; approximate 95% CI **[0.61, 3.39]** minutes. There's evidence of a positive population effect under the test assumptions; deployment still depends on guardrails and business value.

Q14. Why is practical significance different from statistical significance?

Interview answer: With huge samples, a tiny effect can be statistically significant while having negligible user or business value; conversely, a valuable effect may be missed by an underpowered study. Report absolute lift, relative lift, confidence interval, expected aggregate impact, cost, and guardrails. A p-value is not a product strategy.

Q15. What is the minimum detectable effect (MDE)?

Interview answer: MDE is the effect magnitude used when planning a test's target power at a specified alpha and variability. Smaller MDEs require more data. If the business only cares about a 2-minute increase, power the experiment for about that effect, rather than pretending a 0.05-minute change can be reliably detected with the same sample size.

Q16. How do you approximate sample size for a two-group mean comparison?

Interview answer: Under equal independent group sizes, a common Normal-based planning approximation is n per group ≈ $2(z_{1-\alpha/2}+z_{1-\beta})^2\sigma^2/\Delta^2$ for a two-sided test. With α=0.05, power 80%, σ=20 minutes, and target Δ=2, $n \approx 2(1.96+0.84)^2(400)/(4) \approx$ **1,568 per group**, before operational adjustments.

$$n/\text{group} \approx 2 \times (z\alpha + z\text{power})^2 \times \sigma^2 / \Delta^2 \quad [z\alpha = 1.96 \text{ for two-sided } 5\%]$$

Q17. What is the multiple testing problem?

Interview answer: Testing many true null hypotheses gives multiple opportunities to find an apparently significant result by chance. With 20 independent true-null tests, each having exact 5% rejection rate, P(at least one false positive)=$1-0.95^{20} \approx$ **64.15%**. Correlated metrics change that exact calculation but do not eliminate the problem.

Q18. Bonferroni vs Holm vs Benjamini–Hochberg?

Interview answer: Bonferroni controls family-wise error rate (FWER) by testing each of m hypotheses against α/m; it works under arbitrary dependence but can be conservative. Holm is a step-down FWER procedure that is at least as powerful as Bonferroni. Benjamini–Hochberg targets false discovery rate (expected false-discovery proportion under suitable assumptions), often useful for exploratory metric families. Pick the error criterion first.

Q19. Why does peeking at p-values cause trouble?

Interview answer: In an ordinary fixed-horizon test, checking accumulating results repeatedly and stopping at the first $p<0.05$ creates extra opportunities for false rejection. That can inflate the actual false-positive rate. Predefine sample size and stopping rule, or use an appropriate sequential method with valid error control.

Q20. What is Sample Ratio Mismatch (SRM), and why does it matter?

Interview answer: SRM is a statistically suspicious mismatch between observed assignment counts and the intended randomization ratio. If a 50/50 experiment yields 600 in A and 400 in B, investigate bucketing, eligibility, logging, bot filters, redirects, or missing observations. SRM is a diagnostic; don't interpret treatment lift confidently before addressing its cause.

Q21. What are A/A tests and guardrail metrics?

Interview answer: An A/A test assigns users to nominally identical experiences to validate randomization, logging, and expected false-positive behavior over repetitions; one nonsignificant A/A result proves little. Guardrails track unintended harm (crashes, latency, retention, complaint rate, revenue, user satisfaction). An increase in watch time may be unacceptable if trust or safety metrics deteriorate.

Q22. What do you do if results are insignificant but the confidence interval is wide?

Interview answer: Don't claim the algorithms are equivalent. A wide interval may include harmful and valuable effects, indicating insufficient precision. Check whether sample size, noise, experiment duration, and quality were adequate. Distinguish 'no evidence of difference' from 'evidence of no practically important difference'; the latter needs a design such as equivalence testing with prespecified margins.

Q23. What is a unit-of-analysis error in A/B testing?

Interview answer: Randomize users but analyze millions of individual video views as independent observations, ignoring within-user dependence. This typically underestimates the Standard Error and inflates significance. Aggregate to the user level or use an appropriate cluster-aware inference method. The unit of randomization and the unit of statistical independence must be considered together.

Q24. What is novelty effect, and how can experiment duration bias results?

Interview answer: A new interface or recommendation strategy may temporarily change behavior until users adapt. Traffic can also vary by weekday, season, launch campaign, and outages. Plan enough duration to capture a representative behavior cycle, examine predeclared segments, and avoid treating an initial spike as permanent lift.

Q25. What if the primary metric wins but a guardrail gets worse?

Interview answer: A statistically positive primary outcome is not sufficient to ship. Quantify the trade-off, check guardrail uncertainty and predeclared thresholds, diagnose affected segments, and consider controlled rollout or iteration. For a recommendation system, watch-time gains that worsen satisfaction or safety may be unacceptable.

Worked A/B case — narrate this in under 90 seconds

Question: You ran a 50/50 randomized test with 1,600 users per group. New recommendations raised observed mean watch time from 42 to 44 minutes; sample SD is 20 in both groups. What do you recommend?

1. **Estimand:** Difference in mean daily watch time per randomized user, $\delta = \mu_B - \mu_A$.
2. **Observed estimate:** +2 minutes, or +4.76% versus A's 42-minute mean.
3. **Uncertainty:** SE ≈0.707; approximate 95% CI ≈[0.61,3.39].
4. **Evidence:** Welch $t \approx 2.83$, two-sided $p \approx 0.0047$, significant at predeclared $\alpha = 0.05$.
5. **Validity:** Check SRM, exposure/logging, user-level independence, missingness, duration, concurrent launches.
6. **Decision:** Consider gradual rollout if guardrails and practical value are acceptable; monitor outcome stability and affected segments.

Senior follow-up: The CI excludes zero, but does it prove every user benefits? **No.** It concerns an average effect, not individual treatment effects or subgroup guarantees.

CROSS-CHAPTER — Tough Interview Scenarios

Use these to practice a complete answer. Start with the decision, then explain the assumptions and evidence.

Case A. The average rises, but most users are worse off

Watch time increases from mean 42 to 45 minutes, while median falls from 30 to 25 minutes. **Strong answer:** The upper tail may have grown while a typical user's experience declined. Inspect quantiles, distribution shifts, active-user mix, retention and satisfaction. Average improvement alone doesn't guarantee broadly positive impact.

Case B. You have 5 million rows but only 20,000 users

Strong answer: Identify user as the possible clustering/randomization unit; user events are correlated. Don't divide an event-level SD by √(5 million) as though events are independent. Aggregate per user or estimate cluster-robust uncertainty.

Case C. A model flags 1% prevalence incidents with 90% sensitivity and 5% FPR

Strong answer: Use Bayes/base rates. PPV = .009/(.009+.0495)≈15.4%. Discuss operational false-alert load and threshold choice, not only ROC-AUC or sensitivity.

Case D. A/B test produces p=0.049 with only one day of data

Strong answer: Ask whether sample size, stopping plan, metric, exposure period, and multiple testing were prespecified. A p-value just below 0.05 isn't a confidence guarantee or launch order. Report effect magnitude and interval and verify data integrity.

Case E. The t-test says significant, but the distribution is extremely skewed

Strong answer: With enough independent users, the sampling distribution of the mean can still be approximately Normal, but heavy tails may slow CLT convergence and distort variance estimates. Inspect outliers, zero inflation, leverage, group imbalance, and use justified robust/cluster-aware or randomization-based alternatives.

Case F. 20 product metrics, 1 with p=0.03

Strong answer: Was there one prespecified primary metric or were all 20 confirmatory? If all are tested in one family, consider multiplicity control. Bonferroni cutoff 0.05/20=0.0025 means 0.03 doesn't pass. Check for metric cherry-picking and correlated metrics.

Case G. No significant difference; PM says old and new are equal

Strong answer: Failure to reject is not proof of equality. Inspect the CI and its relation to a prespecified smallest practically important difference. Consider an equivalence or noninferiority design if that is the actual business question.

Case H. Treatment group is 55% of assigned traffic instead of 50%

Strong answer: Run an SRM diagnostic before interpretation; investigate randomization and logging. Sample ratio mismatch can signal a systematic flaw affecting the causal estimate.

FORMULA SHEET — The One-page Whiteboard Reference

Topic	Formula / rule
Mean	$\bar{x} = \Sigma x_i/n$
Weighted mean	$\Sigma w_i x_i/\Sigma w_i$
Population variance	$\sigma^2 = \Sigma(x_i-\mu)^2/N$
Sample variance	$s^2 = \Sigma(x_i-\bar{x})^2/(n-1)$
Standardization	$z = (x-\mu)/\sigma$
Covariance	$Cov(X,Y)=E[XY]-E[X]E[Y]$
Pearson correlation	$\rho=Cov(X,Y)/(\sigma_x\sigma_y)$
Probability union	$P(A\cup B)=P(A)+P(B)-P(A\cap B)$
Conditional probability	$P(A\mid B)=P(A\cap B)/P(B)$
Bayes' theorem	$P(H\mid E)=P(E\mid H)P(H)/P(E)$
Expected value	$E[X]=\Sigma x\cdot P(X=x)$ (discrete)
Variance identity	$Var(X)=E[X^2]-E[X]^2$
Sum variance	$Var(X+Y)=Var(X)+Var(Y)+2Cov(X,Y)$
Bernoulli	$E[X]=p$; $Var(X)=p(1-p)$
Binomial	$P(X=k)=C(n,k)p^k(1-p)^{n-k}$
Poisson	$P(X=k)=e^{-}\lambda\ \lambda^k/k!$; $E[X]=Var(X)=\lambda$
Exponential survival	$P(T>t)=e^{-}\lambda^t$; $E[T]=1/\lambda$
Standard Error of mean	$SE=\sigma/\sqrt{n}$; estimated $SE=s/\sqrt{n}$
Sampling error	$\bar{x}-\mu$
Mean 95% CI (Normal approx.)	$\bar{x}\pm1.96\cdot SE$
Margin of error	$MOE=zcrit\cdot SE$
Required n for mean CI	$n=(zcrit\cdot\sigma/MOE)^2$, round up
Standardized mean difference	$Z=$(estimate−null reference)$/SE$
Welch SE for difference	$\sqrt{(sA^2/nA+sB^2/nB)}$
Statistical power	$1-\beta$
n/group for A/B (approx.)	$2(z\alpha+zpower)^2\sigma^2/\Delta^2$
Multiple independent tests, all null	$P(\geq1\text{ false positive})=1-(1-\alpha)^m$
Bonferroni cutoff	α/m

Notation note: These are standard forms under their stated assumptions; they are not universal replacements for design-aware variance estimation. In a two-sided 5% test, $z\alpha$ denotes 1.96 in the A/B planning approximation above.

COMMON INTERVIEW TRAPS — 25 Rapid Corrections

1. **'Median is better than mean.'** It depends on the target: typical user versus total-impact average.
2. **'Outlier = error.'** Some extremes are genuine; investigate first.
3. **'Use n−1 for all variance.'** Population variance and sample estimator are different quantities.
4. **'Zero correlation means independence.'** Generally false; nonlinear dependence can remain.
5. **'Correlation proves causation.'** It doesn't.
6. **'A 90th percentile means 90% have exactly that value.'** It is a threshold/rank summary.
7. **'Independent and mutually exclusive mean the same.'** They are different.
8. **'P(A|B)=P(B|A).'** Bayes' theorem shows why generally not.
9. **'High recall always gives high precision.'** Base rate and FPR matter.
10. **'E[X] must be an attainable outcome.'** Expected value can fall between outcomes.
11. **'A PDF is a probability at a point.'** It's density; integrate over intervals.
12. **'Poisson fits any count.'** Check rate stability, independence, and dispersion.
13. **'All distributions are Normal with enough data.'** CLT concerns sample means, not raw observations.
14. **'SD and SE are interchangeable.'** They quantify different variability.
15. **'SE = actual observed error.'** Not necessarily.
16. **'Large n guarantees unbiasedness.'** Large biased samples can be precisely wrong.
17. **'95% CI means 95% chance this fixed interval contains μ.'** Not standard frequentist interpretation.
18. **'Narrow CI proves correctness.'** Sampling and model assumptions can fail.
19. **'p=0.01 means 99% probability the alternative is true.'** Incorrect interpretation.
20. **'p-value is effect size.'** No; report estimated lift.
21. **'Not significant means no effect.'** Might be underpowered or imprecise.
22. **'Type I = false negative.'** Type I is false positive; Type II is false negative.
23. **'80% power means 80% chance the effect exists.'** It's a detection rate under a specified true effect.
24. **'n<30 always t, n≥30 always Z.'** Test assumptions and variance estimation decide.
25. **'We can stop when p first hits 0.05.'** Not with an unadjusted fixed-horizon test.

MOCK INTERVIEW — Two Short Rounds

Round 1: Fundamentals (15 minutes)

Prompt 1: Mean 48, median 30, p99 240. Describe the watch-time distribution and two follow-up analyses.

Prompt 2: A rare class occurs in 1% of sessions. Explain why 90% recall may produce poor precision.

Prompt 3: An event count has mean 3 and variance 30. Is a plain Poisson model plausible? Why?

Prompt 4: If individual watch times are skewed, can an average have an approximately Normal sampling distribution?

Prompt 5: Distinguish a 95% confidence interval from a 95% prediction interval.

Prompt 6: What does a p-value of 0.04 say — and not say?

Strong-answer criteria: Correct definition; a numerical or product example; major assumption; at least one failure mode.

Round 2: Applied Experimentation (20 minutes)

Prompt 1: Plan a YouTube A/B test to detect a 2-minute watch-time increase. Define estimand, unit, alpha, power, MDE, metric, and guardrails.

Prompt 2: B improves mean by 2 minutes, p=0.0047, 95% CI [0.61,3.39]. Explain the result to a PM without promising a launch.

Prompt 3: You ran 30 metrics and found three p-values just below 0.05. How do you interpret them?

Prompt 4: Assignment intended 50/50, but observed 60/40. What's your first action?

Prompt 5: PM monitored daily and stopped at first $p<0.05$. Is the reported p-value valid for that procedure?

Prompt 6: The overall lift is positive but new users lose watch time. How do you analyze the risk without post-hoc cherry-picking?

Strong-answer criteria: Clear causal design; correct error-rate handling; appropriate unit of analysis; uncertainty and practical impact; data-quality diagnostics; explicit rollout decision framework.

FINAL REVISION PLAN — Three Passes

Pass 1 — 30 minutes: Read the Fast Recall tables for all five chapters and reproduce the Formula Sheet from memory. Focus on distinguishing terms, not memorizing every symbol.

Pass 2 — 60 minutes: Answer the Core Interview Questions out loud without looking; inspect any shaky answers. Prioritize your previously missed areas: p-values vs effect size, Type I vs II, β versus power, sample-size/power direction, independent vs paired tests, and optional stopping.

Pass 3 — 45 minutes: Do the eight Cross-chapter Cases and both Mock Interview rounds. For each, say the practical decision first, then the evidence and assumptions. Time yourself.

Self-scorecard

Dimension	Score 0–2	What 2 looks like
Concept clarity	__ / 2	Definition and intuition are both correct
Math fluency	__ / 2	Formula used with proper assumptions
Assumptions	__ / 2	Notices sampling, independence, and distribution limits
Production thinking	__ / 2	Spots logging, SRM, metrics, and rollout risk
Communication	__ / 2	Clear answer within 60–90 seconds
Total	**__ / 10**	**8+ = confident interview readiness for this topic**

Closing perspective

You do not need to memorize 100 disconnected formulas. Interviewers are evaluating whether you can go from **data → uncertainty → inference → decision** without making a false claim. State assumptions, choose the right unit, explain the practical meaning, and be willing to say *we do not have enough evidence yet*.

End of Data Science Foundations — Interview Field Guide.